

**Rincon's Bio Weaver Tool:
From Disparate Data to Actionable Analysis**

Technical Documentation

A Capstone Project submitted in partial satisfaction of the requirements for the degree
of Master of Environmental Data Science for the
Bren School of Environmental Science & Management

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Abstract

Rincon Consultants, Inc. is a leading environmental consulting firm across California, where biologists conduct reviews under the California Environmental Quality Act (CEQA) to evaluate how proposed projects may impact special-status species and determine their potential to occur within project areas. Currently, Rincon's biologists rely on a fragmented workflow involving manual inputs, search queries, and data joins to compile species occurrence data from sources like the California Natural Diversity Database (CNDDB) and the California Native Plant Society's Rare Plant Inventory. This process introduces inefficiencies, data inconsistencies, and reproducibility issues that can obscure important ecological relationships. The Bio Weaver tool addresses this data management and analysis challenge through an automated data processing workflow that is integrated with a dynamic platform, where biologists can interact with project-specific data in the form of maps, charts, and tables that can then be exported for their regulatory reporting needs. This provides Rincon with a streamlined process for data management and analysis, which will allow biologists to spend less time wrangling data and more time in the field, in turn strengthening long-term resource conservation and ecological resilience across California.

Executive Summary

California's diverse ecosystems are under mounting pressure from urban development, habitat fragmentation, and accelerating climate change (Cashwell, 2025). These forces drive shifts in species distributions and increase the vulnerability of already at-risk plants and wildlife (Cashwell, 2025). As environmental stewards, Rincon's biologists conduct biological reviews under the California Environmental Quality Act (CEQA) to evaluate how proposed projects might affect special-status and culturally significant species. However, the complexity of today's environmental landscape demands a more comprehensive and connected approach to understanding species' occurrences. Existing datasets such as the California Natural Diversity Database (CNDDDB) and the California Native Plant Society's Rare Plant Inventory offer vital insights but operate in isolation. Integrating these resources provides an opportunity to reveal ecological relationships that are often overlooked when data remain siloed. By connecting and visualizing these datasets within a unified analytical framework, Rincon's team can better anticipate where sensitive species may occur and promote proactive planning that safeguards biodiversity. This effort represents a step toward more adaptive, data-informed conservation practices that strengthen long-term resource management and ecological resilience across California.

Rincon's current approach to compiling species occurrence data and creating reports is largely manual, relying on biologists to go to individual data websites, filter data based on project-specific parameters, download files, and conduct data joins using an in-house tool. The end goal of this data pipeline is to create a preliminary Potential To Occur (PTO) table, which compiles species occurrence data from the many databases. Biologists can then add their assessments of how likely these species are to occur based upon the available data, as well as field observations at the project site. This lengthy and complex process introduces inefficiencies, data inconsistencies, and reproducibility issues that can obscure important ecological relationships.

Our project will address this data management challenge by designing an automated, reproducible data workflow that integrates data from multiple sources into a dynamic platform, where users can easily query these databases based on project-specific requirements, retrieve a preliminary PTO table, and explore resulting maps and charts to make informed potential-to-occur determinations in one user-friendly application. Overall, we will provide Rincon with a scalable, efficient system that enhances data integrity, reduces redundancy, and empowers Rincon's biologists to focus on ecological interpretation rather than data wrangling.

The final product will be handed over to Rincon in the form of three deliverables:

- **Reproducible Data Processing Workflow:** The team will develop a reproducible workflow to standardize and process all biological occurrence datasets used by Rincon biologists.
- **GitHub Repository:** The capstone team will deliver a private GitHub repository containing the full codebase and project architecture. This repository will serve as the technical home of

the tool and will allow Rincon to maintain, update, or extend the system.

- **Interactive Application:** The capstone team will deliver a centralized, user-friendly application that accepts biologist-defined parameters, allows exploration of resulting data, and delivers ready-to-use results to support Rincon's biological review workflows.

This capstone project goes beyond just a simple application tool. It is clear that there is a growing need for environmental consulting workflows to become more scalable, more consistent, and more accessible, especially as biological data and projects become more complex. The Bio Weaver is just one example we have set of how we can use integrated data tools to move workflows in that direction.

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1. Approach

To create a robust tool that can be iterated upon and maintained post-project completion, the Bio Weaver tool is made up of three development phases: backend, frontend, and packaging. First, we conduct a full download of the California Native Plant Society data and obtain a randomized version of the California Natural Diversity Database. We then formulate our backend to act on this data, made up of Python (PyPI, 2025) scripts containing functions for geometry handling, species filtering, and Potential to Occur table generation. We then design the frontend using Streamlit (Snowflake Inc., 2026) to allow for user inputs and project parameters to be specified, and then a page to visualize the outputs in the form of a map, charts, and the formatted Potential to Occur table that can be exported to Word format. Lastly, we conduct brief user testing and finalize the documentation within the GitHub repository, and package the tool for a smooth transition to independent use by Rincon Consultants, Inc.

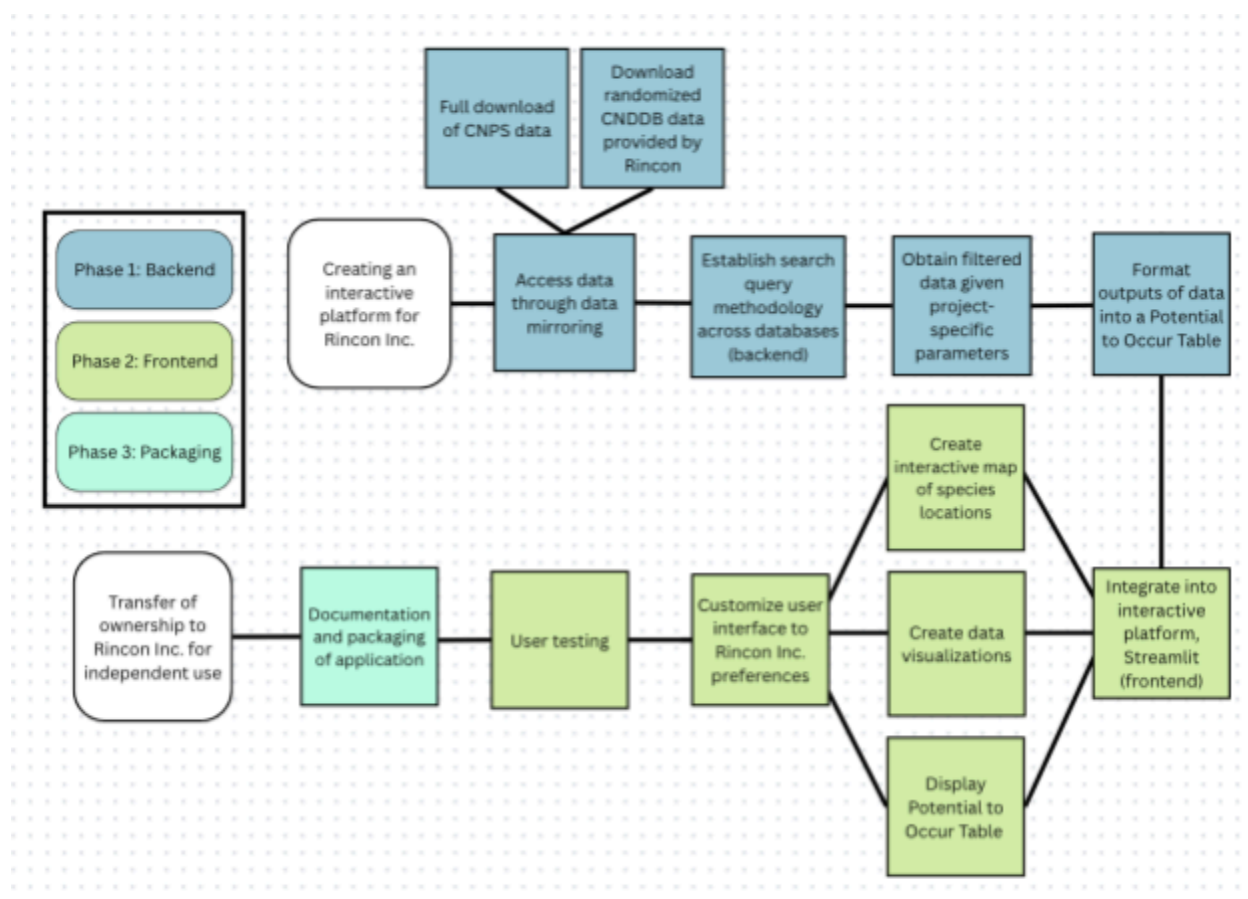


Figure 1: Workflow diagram for the creation of the Bio Weaver Tool, divided into three main phases: backend, frontend, and packaging for the client.

2. Methods

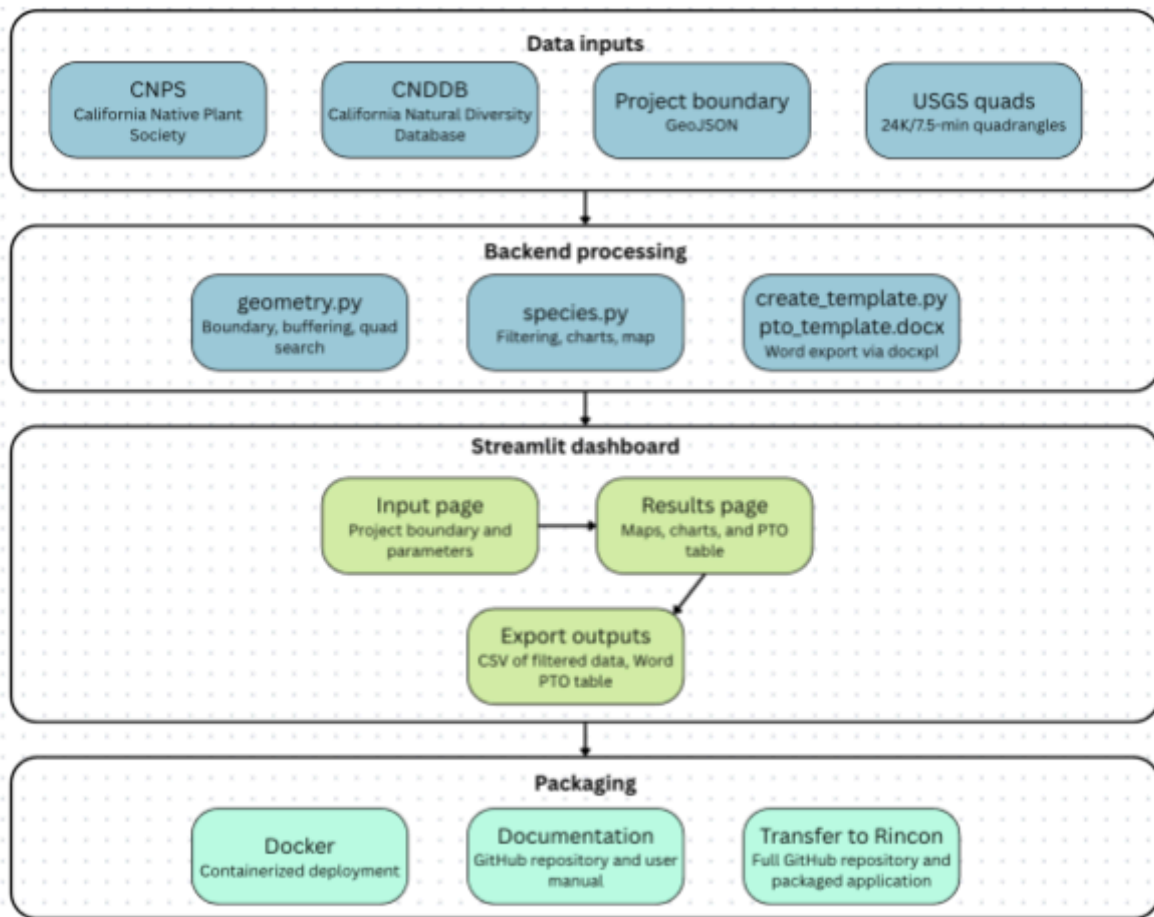


Figure 2: Methods overview of the Bio Weaver tool, structured into data inputs, backend processing, Streamlit application, and packaging.

This approach serves to simplify and improve the existing workflow used by biologists to achieve the same outputs, where they search separate databases, download files, cross-validate sources, conduct manual data cleaning, and obtain spatial layers from a geospatial team.

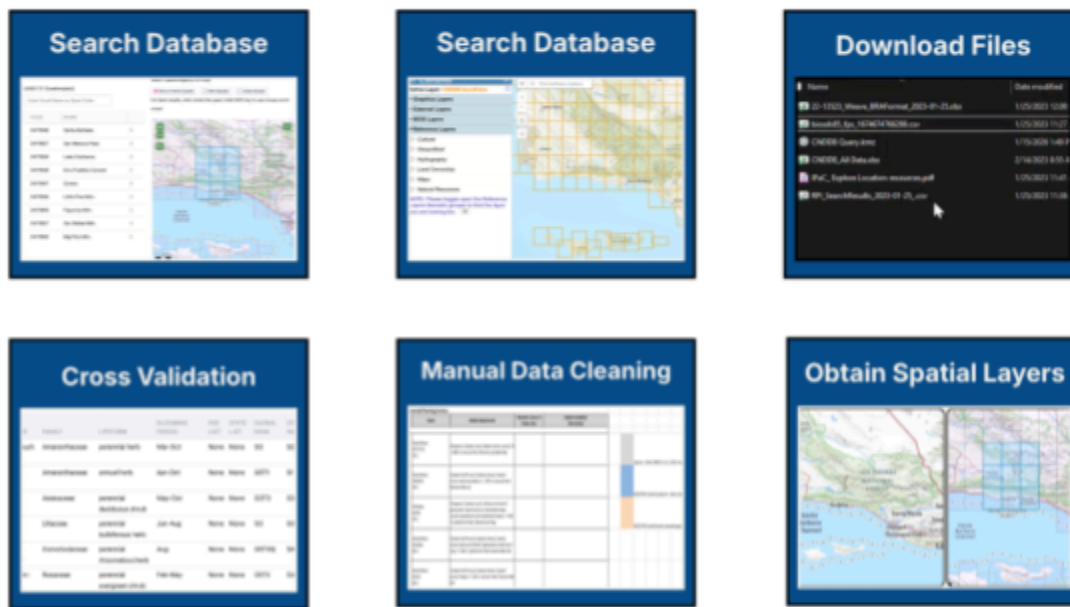


Figure 3: Existing Rincon Workflow: Illustration of the current biologist workflow at Rincon, showing how species data is gathered, validated, and manually processed across multiple tools and hand-offs.

2.1. Data

The Bio Weaver Tool will utilize three key biological databases: the California Natural Diversity Database (CNDDDB), the California Spotted Owl Observation Data, and the California Native Plant Society (CNPS) Database. These databases are used regularly by Rincon biologists to understand species occurrences in relation to specific biological project evaluations.

The information we are focused on drawing from these databases includes species name, common name, status, and various habitat requirements. These specific attributes are used to generate project-specific species lists, summaries, and Potential to Occur tables. The biological databases also include USGS quadrangle attributes, which are used as the spatial information to filter the species within the databases. These quadrangles are only utilized within the backend to conduct the spatial search and are not displayed within the PTO table or application.

The tool intakes various project-specific spatial boundaries in the form of GeoJSON files, which define the spatial area of the biological evaluation. A key dataset driving the methodology of the Bio Weaver functionality is the California Statewide Index of USGS 24K/7.5-Minute Quadrangles, a publicly available dataset accessed via ArcGIS. This essential dataset provides a comprehensive list of all quadrangles within the state of California. As part of the evaluation workflow, Rincon biologists may implement a 9-quad buffer on the project boundary. In other words, a project area can include 9 USGS 24K/7.5-Minute Quadrangles that intersect/overlap the project boundary. Therefore, each project boundary is cross-referenced against the index of quadrangles to identify which

quadrangles intersect the project area, forming the spatial basis for querying species occurrence data and mapping purposes.

2.1.1. Sources

The following datasets are inputs to the Bio Weaver Tool:

Table 1. Source Datasets and Spatial Formats.

Data for Bio Weaver Tool	
<i>Inputs</i>	<i>Source</i>
Project-Specific Boundaries Example project GeoJSON files.	Rincon Consultants, Inc.
California Natural Diversity Database (CNDDDB) Statewide inventory of rare, threatened, endangered, or other special status species and natural communities.	California Department of Fish & Wildlife (CDFW)
Spotted Owl Observation Data Survey records and location data for northern and California spotted owls.	https://wildlife.ca.gov/Data/CNDDDB/Maps-and-Data
California Native Plant Society (CNPS) Conservation status of California's rare, threatened, and endangered plant species.	California Native Plant Society (CNPS) https://rareplants.cnps.org/Search/Advanced
California Statewide Index of USGS 24K/7.5-Minute Quadrangles A subset of an index of the new USGS "US Topo" series of maps published as digital documents in 2009 and later.	ArcGIS https://www.arcgis.com/home/item.html?id=2f5ff4c005ee47798a32afb1e4e25624&dataTabView=table#data

2.1.2. Data Management

The first attempt at accessing data involved accessing APIs from the California Department of Fish & Wildlife and the California Native Plant Society; this method was unsuccessful due to incomplete or restricted API services.

As a result, Rincon conducted full dataset downloads for both the CNDDDB and CNPS databases. The CNDDDB dataset was additionally randomized to comply with data privacy and licensing requirements of the California Department of Fish & Wildlife. Therefore, all datasets used during the

development process of the Bio Weaver tool represent static datasets that do not reflect updates made by the original data sources. These datasets are strictly used only for testing and development purposes and will not be used by Rincon at the conclusion of the Capstone project.

To maintain data accuracy, the system design assumes that Rincon will periodically update datasets via manual re-downloads. This update process is external to the Bio Weaver Tool, but it is necessary to ensure that analyses are based on current data.

Datasets are stored locally within a standardized directory structure (bioweaver/data). Updated datasets are expected to replace existing files while maintaining consistent file naming conventions to preserve compatibility with the Bio Weaver Tool, which calls datasets given predefined file paths. This structure supports reliable data access, simplified data replacement, and streamlined integration of databases into the tool's search query pipeline.

Data can easily be replaced by following the same naming conventions within the file structure. Where the California Natural Diversity Database folders and all associated files are named ``cnddb`` (`cnddb.dbf`, `cnddb.prj`, `cnddb.shp`, `cnddb.shx`) and the California Native Plant Society CSV file is named ``cnps.csv``.

2.2. Backend Functionality: Search Query Methodology

The backend functionality of the Bio Weaver Tool is implemented through a series of modular Python scripts that handle data processing, spatial querying, and dataset integration. The search query methodology is the core of the backend system, translating user-defined spatial inputs into structured queries that use functions to retrieve relevant project-specific species occurrence records across biological datasets, leveraging spatial identifiers and standardized data structures.

2.2.1. Geometry

The “geometry.py” script contains functions responsible for processing project-specific spatial inputs and translating them into spatial queries. This module forms a critical component of the backend workflow, as it defines the spatial extent of Rincon biological evaluations and identifies the spatial units used to query biological datasets. Functions within this script can be expanded upon to become more complex, or additional functions can be created within this script to implement more data processing functionality.

Project Boundary Definition

Project-specific spatial boundaries are integrated into the Bio Weaver Tool as a GeoDataFrame from a GeoJSON file. The coordinate reference system (CRS) is standardized to WGS84 (EPSG:4326) to ensure consistency across databases and compatibility with subsequent spatial operations.

Spatial Preprocessing

Once loaded, the boundary geometry may be further processed to support additional project-specific spatial analysis requirements. This includes the optional application of a buffer around the project boundary using the California Albers, EPSG:3310 projected coordinate reference system to ensure accurate distance-based calculations.

Buffer distance options include **2-mile**, **5-mile**, **10-mile** radius, as well as the **9-quadrangle** search area, which expands the search extent to the project's primary quadrangle plus the 8 surrounding USGS quads. The returned buffered project boundary is then transformed to the original WSG84 CRS for consistency throughout the rest of the workflow.

USGS Quadrangle Identification

The California Statewide Index of USGS 24K/7.5-Minute Quadrangles dataset is integrated as a GeoDataFrame and is used to define the spatial units for querying species occurrence data sets. The quadrangle identifiers are first standardized into 7-digit integer quad IDs, which serve as the primary spatial key used to link the spatial data inputs to the biological datasets.

First, a bounding box from the buffered project boundary is created. Then, based on the bounding box, a set of quadrangles that intersect the bounding box is returned. This approach improves computational efficiency by reducing the spatial search operation to a smaller subset of relevant quadrangles rather than performing the quadrangle intersection across the entire California statewide dataset.

Species Query

The resulting set of quad IDs is used to query the biological datasets and obtain relevant project-specific species occurrence data. CNPS species occurrence data is reported at the quadrangle level, rather than at spatial coordinates or geometries. Therefore, quad-based filtering is the primary method for identifying species within a project area.

Given the spatial geometries provided within the CNDDDB dataset, the search query is applied to the CNDDDB dataset using a spatial filter, where species occurrences are identified based on whether they intersect the user-defined buffered areas around the project boundary. The separate functions take into consideration the structural differences between species location identifiers.

2.2.2. Species

The “species.py” script contains functions that create chart summaries of the species occurrences. This module forms another critical component of the backend workflow, as it generates the data visualizations that will be ingested into the frontend of the Bio Weaver Tool.

Data Visualization

To support exploratory analysis and interpretation of species occurrence data, the module includes visualization functions for summarizing spatial and temporal patterns. A bar chart representing the number of occurrences for each species provides an overview of the species prevalence within the project area. Additional functions that produce various charts would be added to this script, with an example provided within the codebase.

Map Visualizations

To provide a spatial understanding of the distribution of species within a project area, a map highlighting the boundary of the project area and the presence of CNDDDB species within the project area. The map is supported through the Python mapping library, Pydeck. This dependency allows for interactive features, such as zooming, scaling across the map, and definitions of species. The script includes a function that creates a general map and a function that creates a map to be displayed in the Streamlit application.

2.2.3. PTO Table Generation

The PTO table is exported as a formatted Word document using `docxtpl`, a templating wrapper around `python-docx` that enables Jinja2-style placeholders in Word files. When an analysis is run on the Streamlit application, the backend retrieves the filtered CNPS and CNDDDB data and iterates over each row to populate the templated text blocks for each species. These fields include scientific and common names, regulatory status, habitat description, elevation range, and blooming period. The data is passed as context to the Word template, which is then rendered and saved as the final output document.

All outputs include:

- CSV files of raw data tables of filtered CNDDDB and CNPS tables
- A Word document containing the formatted PTO table.

Required Scripts for PTO Report Generation

- `create_template.py`

The `create_template` script contains the function responsible for generating the base Word template stored in the `src/` directory. This function is run only once to produce the blank PTO report template. To regenerate the template, run `create_template.py` from the command line. This function only creates the empty template; it does not insert any species data. Data population occurs later through `make_buffer()` using `docxtpl` (Eric Lapouyade, 2015).

- `make_buffer.py`

The `'make_buffer'` script populates the PTO report with the selected search-area results. It retrieves the most recent editable PTO table stored in Streamlit's session state, organizes the species records by taxonomic category, renders them into the Word template using `'docxtpl'`, and returns the completed Word document as bytes. If no table is available, or if rendering fails, the function returns `'None'`.

- `format_data.py`

The `'format_data'` script contains the functions responsible for converting CNPS and CNDDDB source columns into the standardized fields used by both the Streamlit PTO table and the Word export. These helper functions ensure consistent formatting across data sources and output products.

2.3. Frontend Functionality: Application Integration

The Bio Weaver Tool's frontend was built using Streamlit, an open-source Python application framework. This framework was selected due to its lightweight, dynamic, and fast nature, as well as the built-in features that relate directly to the needs of the tool. Streamlit works directly with the Python scripts housed within the repository. These scripts are `'Home.py'`, `'pages/1_Search.py'`, `'pages/2_Results.py'`, and `'pages/3_Table.py'`. Each script corresponds to a page that is rendered within the Bio Weaver Tool application; based on the file structure, Streamlit automatically recognizes them as a multi-page interface. Along with Streamlit-specific functions, these scripts import the functions created within `'geometry.py'` and `'species.py'` to execute the data processing workflow within a user-friendly interface. When run as a Streamlit application, the Bio Weaver Tool:

- **Processes the Workflow:** When the user uploads project GeoJSON files and selects certain project parameters, Streamlit will process the backend functions.
- **Dynamic Updates:** Results are produced on the results page through Streamlit components, including interactive maps, tables, and summary metrics.
- **Error Handling:** The frontend includes response validation. If the inputs cause an error code or retrieve an empty dataset, the application displays a clear message and troubleshooting guidance.

2.4 Packaging and Local Command-Line Deployment with Docker

To simplify deployment, the Bio Weaver Streamlit application is packaged using Docker (Snowflake Inc., 2026). This allows the application to run in a self-contained environment without requiring users to install Python, manage dependencies, or configure environments manually. It standardizes the environments, allowing Rincon biologists to use the Streamlit app on different local computers. The application can then be launched locally from the command line. The Dockerfile is located in the root directory of the repository. Packaging the Bio Weaver Tool within Docker allows Rincon to later deploy the application on their own remote server after the conclusion of the Capstone Project (Docker, 2026).

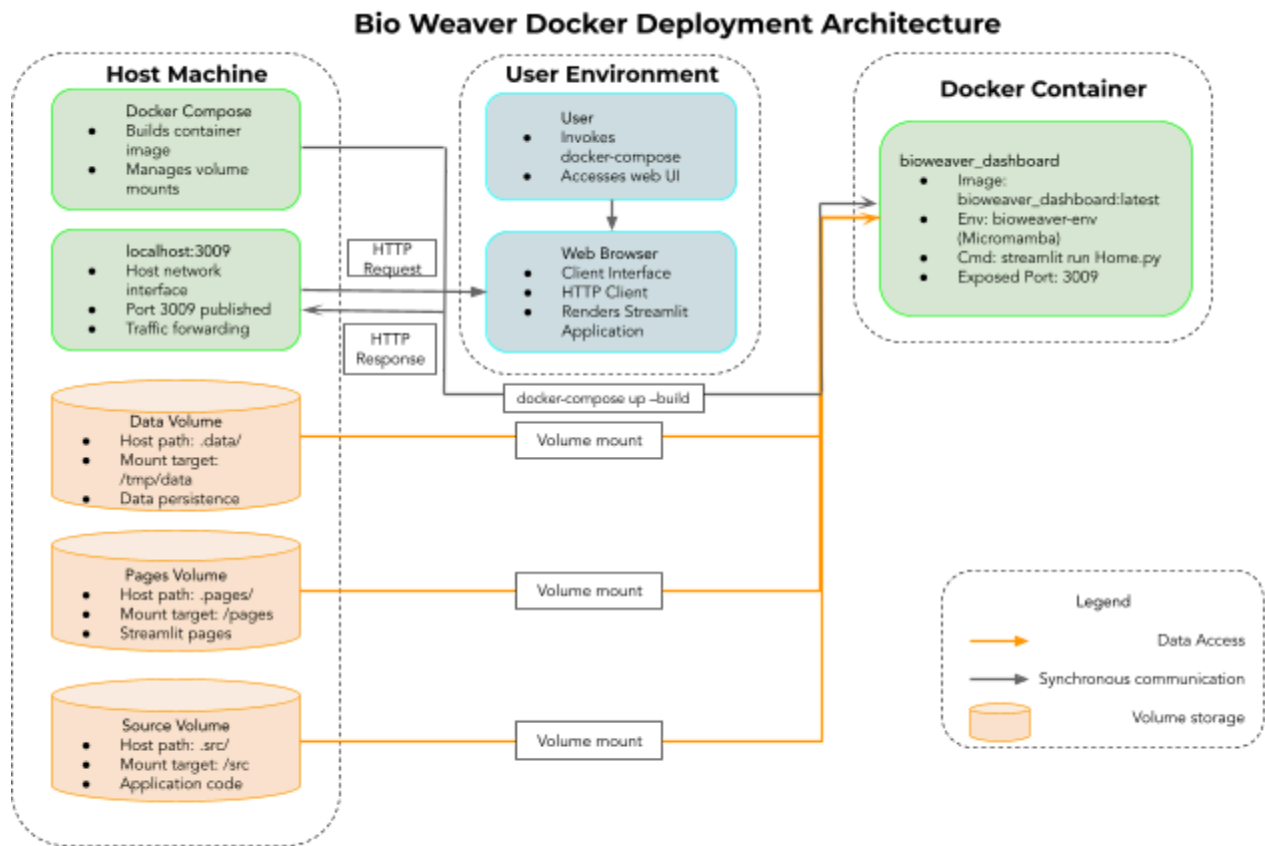


Figure 4. Diagram illustrating how the Bio Weaver application runs inside a single Docker container orchestrated by Docker Compose.

Docker Configuration Files

The Bio Weaver Tool includes two configuration files that define the containerized environment:

- *Dockerfile*: specifies the `micromamba` base image, installs dependencies from *environment.yml*, copies the application code, exposes port 3009, and launches the Streamlit application.
- *docker-compose.yml*: builds the container image, mounts the local *data/*, *pages/*, and *src/* directories into the container, and publishes port 3009 for browser access.

These files are located in the root directory of the repository and provide a reproducible environment for running the application locally. More detailed instructions for building and running the Dockerized application are provided in Section 4.1. This section focuses on ensuring that Docker is installed and configured correctly on the user's machine.

3. Product Description

The platform serves as a centralized interface that accepts biologist-defined project parameters, allows exploration of resulting data through maps, tables, and charts, and delivers ready-to-use reports to support Rincon’s biological review workflows.

3.1. Intended Users

The primary users of the Bio Weaver Tool are biologists tasked with identifying the potential occurrence of species within a project footprint. These users are familiar with the CNDDDB and CNPS datasets. Biologists use these data to explore, analyze, and retrieve data on the measures they prioritize. Their goal is to obtain data on species occurrences, note field observations, habitat descriptions, and nomenclature to support potential-to-occur determinations that serve as the legal justification in biological assessments.

3.2. Reproducible Search Query Pipeline

The search query pipeline exists within Python scripts, which serve as the backend when users select parameters in the interface. These functions can be used independently of the platform as well if desired. The purpose of the Reproducible Search Query Pipeline is to provide Rincon Consultants Inc. with a foundational codebase pipeline where the company is able to build upon and improve the tool after the completion of the MEDS capstone project.

3.3. Interactive Application

The interactive application consists of a welcome page, a search page, and a results page. On the welcome page, there is an introduction to the tool and a description of how to navigate through each page. The user can then click a “Start Your Analysis” button, which directs the user to the search page. On the search page, the user can upload a project-specific boundary (GeoJSON) and select their desired parameters for the project. This includes selecting radius criteria, in the form of 2-mile, 5-mile, or 10-mile buffers around the project boundary or a 9-quad search. Once these parameters are selected, the user can click a “Go to Next Page: Results Page” button, which sends the selected parameters to the backend functions to return the filtered data and create the resulting map, visualizations, and PTO table and simultaneously directs the user to the results tab, where the results will be displayed. An interactive map shows the locations of the species occurrences within the search area with orange shapes representing animals and green shapes representing plants, and allows the user to hover over each to display a pop-up of attributes. A chart of the number of species occurrences is located directly beneath the map. The species results from each database (CNDDDB and CNPS, respectively) are shown, and an option to rename and download the individual raw CSV files. The user can then click a “Go to Next Page: Potential to Occur Table Page” to explore the PTO table that was generated. The user can edit the final two columns, with the potential to occur with a drop-down menu, and habitat suitability. The count of assigned animals updates at the top with each

edit saved. Once the table is edited as desired and the edits are saved, there is the option to rename and export it as a Word file from the platform. The PTO table can also be edited in Word after export.

3.4. Testing

The testing of the deliverables comes in two forms: unit testing and user testing. Key functions within the reproducible search query pipeline have unit tests implemented within “test_geometry.py”, which serve as checks that the functions return the expected outputs. Tests that the environment, with specific packages, versions, and libraries, is housed within the “test_environment.py”. Throughout the project, a common problem we encountered involved spatial libraries and coordinate reference systems. Therefore, several tests to address these issues were created within the “confest.py” and “test_to_crs.py” scripts. The “test_pto.py” tests the generation of the PTO report template and the integration of data into the template.

For testing the functionality of the interface by users, a user testing meeting was conducted during which the application was tested on a Rincon computer with the actual CNDDDB data.

3.5. GitHub Repository

The final GitHub repository (<https://github.com/bioweavers/bioweaver-tool>) contains the entire functional codebase of the platform, including the backend Python scripts and the frontend Streamlit files. It includes an in-depth README file that documents the structure of the repository and general information about how to use the platform.

The preliminary GitHub repository (<https://github.com/bioweavers/bioweavers-mvp>) contains the entire codebase used during the development phase of the project. It contains all the scripts found in the final GitHub repository, including additional resources such as a test demo notebook, test datasets, and the full commit history of the Bio Weaver Tool. The preliminary repository is available for Rincon to reference; however, the final GitHub repository serves as the foundation for future development.

3.6. Performance and Limitations

The tool requires continuous updates of the underlying data, as described in more detail in section 4.1. The current tool maintains state between pages within the project instance, allowing the biologist to move between the inputs, results, and table pages as needed within one project analysis. However, the memory of each project is not sustained within the Streamlit application and must be rerun to replicate results.

4. User Manual

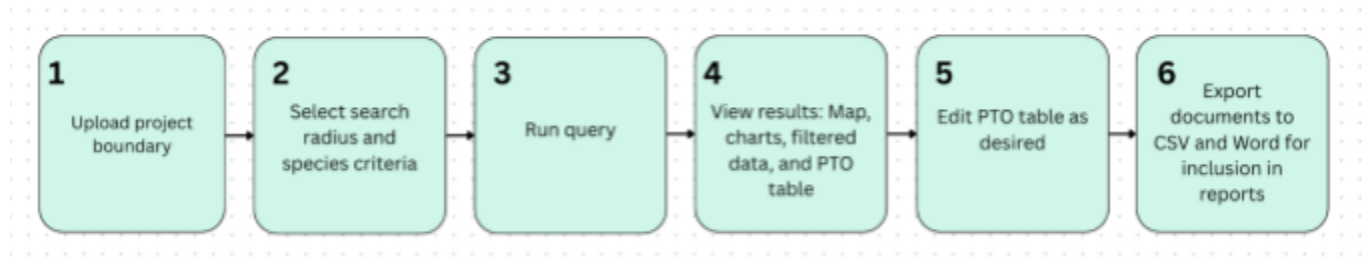


Figure 3: Full user experience of the Bio Weaver tool for biologists

The Bio Weaver Tool is a Streamlit-based web application designed to support species occurrence analysis for environmental consulting workflows. It allows users to upload a project boundary, apply spatial and biological filters, and generate a set of outputs including mapped occurrences, summary statistics, and a formatted report-ready table.

The tool streamlines early-stage biological assessments by reducing the time required to gather and organize species occurrence data from multiple sources. The interface is structured as a simple workflow:

1. Upload the project boundary
2. Select search and species criteria
3. Run query
4. Review results
5. Edit PTO table
6. Export documents to CSV and Word

4.1. Platform Deployment

The application is designed to run locally from the command line, allowing the client to deploy and use the tool without requiring a hosted server.

Development Environment Setup (One-Time Setup)

1. Install VS Code (optional but recommended): <https://code.visualstudio.com/>
2. Install Mamba:
[Mamba Installation — documentation](#)
3. Clone the repository from Git. In the terminal, run:

```
git clone https://github.com/bioweavers/bioweaver-tool.git
cd bioweaver-tool
```
4. Create a new environment. In the terminal, run:

```
mamba env create -f environment.yml
```
5. Activate the environment. In the terminal, run:

```
mamba activate bioweaver-env
```

Running the Application Locally

1. Navigate to the root folder.
cd bioweaver-tool
2. In the terminal run:
streamlit run Home.py
3. Run the application in a browser on your local machine
http://localhost:8501
4. Close the application when finished with analysis. In the terminal, run:
Ctrl C

Important: Do not run multiple instances of the app at the same time. If you see a “port already in use error”, stop any running Streamlit processes.

Installing Docker

To set up Docker, first install Docker Desktop for your operating system:

<https://docs.docker.com/desktop/>

If the operating system is a Windows computer, make sure to have the latest version of Windows Subsystem for Linux (WSL) installed. First, check that what the existing version of WSL is in the operating computer by running the following in the terminal:

```
wsl --version
```

If WSL is not installed or requires an update, run:

```
wsl --install
```

```
wsl --update
```

Once Docker Desktop has been installed, verify the installation by running:

```
`docker --version`
```

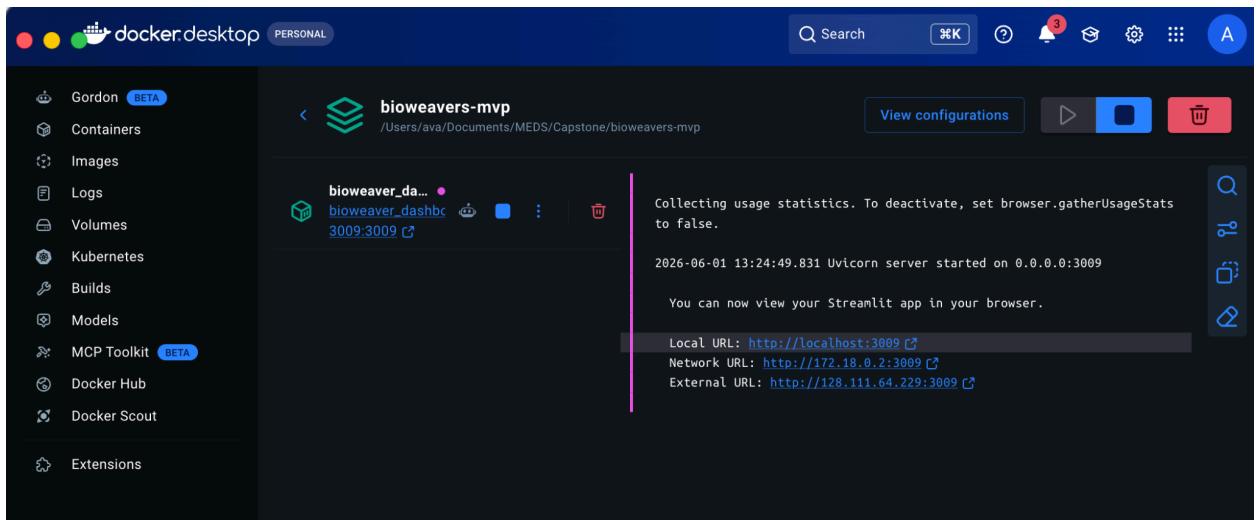
Running the Application with Docker

1. **Prerequisites:** Docker Desktop is installed and running
2. **First-time setup:**
 - a. Open a terminal and navigate to the project root directory. Confirm you’re in the right place by running this command:
pwd
The file path returned should end in **/bioweaver-tool**
 - b. In the terminal, run the command to build and start the container:
docker-compose up --build

3. **Opening the application:** Open the application in a browser on your local machine through the terminal: <http://localhost:3009>

```
[+] up 2/2
✓ Image bioweaver_dashboard:latest Built
✓ Container bioweavers-mvp-bioweaver_dashboard-1 Recreated
Attaching to bioweaver_dashboard-1
bioweaver_dashboard-1 | Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.
bioweaver_dashboard-1 | 2026-06-01 13:24:49.831 Uvicorn server started on 0.0.0.0:3009
bioweaver_dashboard-1 | You can now view your Streamlit app in your browser.
bioweaver_dashboard-1 | Local URL: http://localhost:3009
bioweaver_dashboard-1 | Network URL: http://172.18.0.2:3009
bioweaver_dashboard-1 | External URL: http://128.111.64.229:3009
bioweaver_dashboard-1 |
View in Docker Desktop View Config Enable Watch Detach
```

- Alternatively, in Docker Desktop, go to the Containers tab, find `bioweaver-tool`, and click the play button. Then click on the local URL: <http://localhost:3009>



4. **Stopping the app:** Stop the Docker container when you are done with the application. In Docker Desktop, click the square stop button next to the `bioweaver-tool`.

Within Docker Desktop, the `bioweaver-tool` container can be stopped and restarted as desired throughout the use of the tool- this is the easiest startup method after the application container is first built.

Data Setup and Updates

Before running the application, ensure that all required datasets are correctly placed and formatted.

Data Directory

All input data must be stored in the data/ folder.

- data/ folder
Include only the updated files in this folder.

```

├── data/
│   ├── cnddb/                # Replace with current CNDDDB data folder as needed
│   └── cnps.csv              # Replace with current CNPS data as needed
├── images/                  # Logos and other assets
├── pages/
│   ├── 1_Search.py          # Dashboard page 1
│   ├── 2_Results.py         # Dashboard page 2
│   └── 3_Table.py           # Dashboard page 3
├── src/
│   ├── create_template.py    # Create PTO Word template
│   ├── format_data.py        # Format data for PTO table
│   ├── geometry.py           # Spatial functions
│   ├── make_buffer.py        # Generate PTO from template
│   ├── species.py            # Species map and chart functions
│   └── pto_template.docx     # Template for PTO table
├── tests/
│   └── fixtures/
├── docker-compose.yml       # Files to create Docker container
├── Dockerfile
├── environment.yml          # Python environment
├── Home.py                  # Dashboard Home page
└── README.md

```

Data Requirements

- Only include the most recent versions of datasets
- Do not rename files unless instructed
- Ensure files follow expected naming conventions:
 - Inside the cnddb data directory:
 - cnddb.prj
 - cnddb.shp
 - cnddb.shx
 - cnddb.dbf
 - cnps.csv

Updates to Codebase

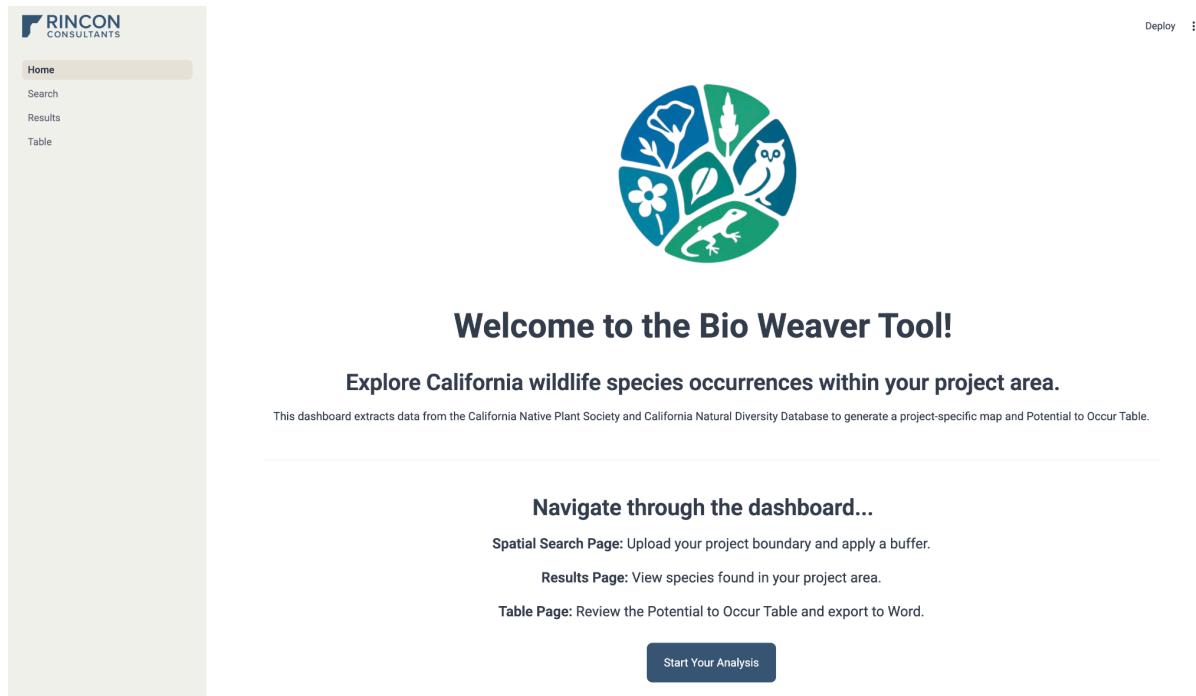
Before making any changes to the Bio Weaver Tool codebase:

1. Run the following terminal command to stop and remove the current container
docker-compose down
2. After making changes to the codebase, rebuild the container so that the Bio Weaver Tool reflects any updates.
 - a. Repeat steps 1-4 of **Running the Application with Docker** to rebuild a new container that reflects all updates to the codebase, **such as updating the CNPS and CNDDDB databases.**

4.2. Getting Started (Step-by-Step Guide)

Step 0: Navigate through the Home Page

- Read through the introduction of the Bio Weaver tool
- Understand the functionality of each page
- Click the “Start Your Analysis” button



Step 1: Upload Project Boundary

- Upload a project boundary file (GeoJSON format)
- The boundary defines the spatial extent of the analysis
- Once uploaded, the boundary is previewed on the map



[Home](#)

[Search](#)

[Results](#)

[Table](#)

Deploy

Spatial Search Page

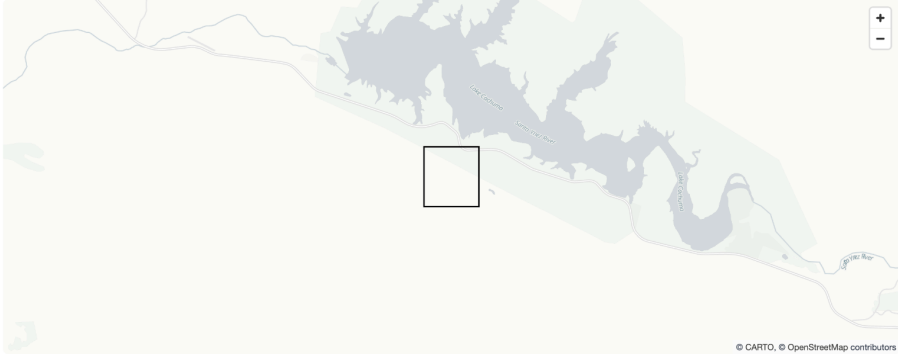
Upload Project Boundary

Please upload the project boundary GeoJSON file.

 lake_cachuma.geojson 327 KB

+

Project Boundary Preview

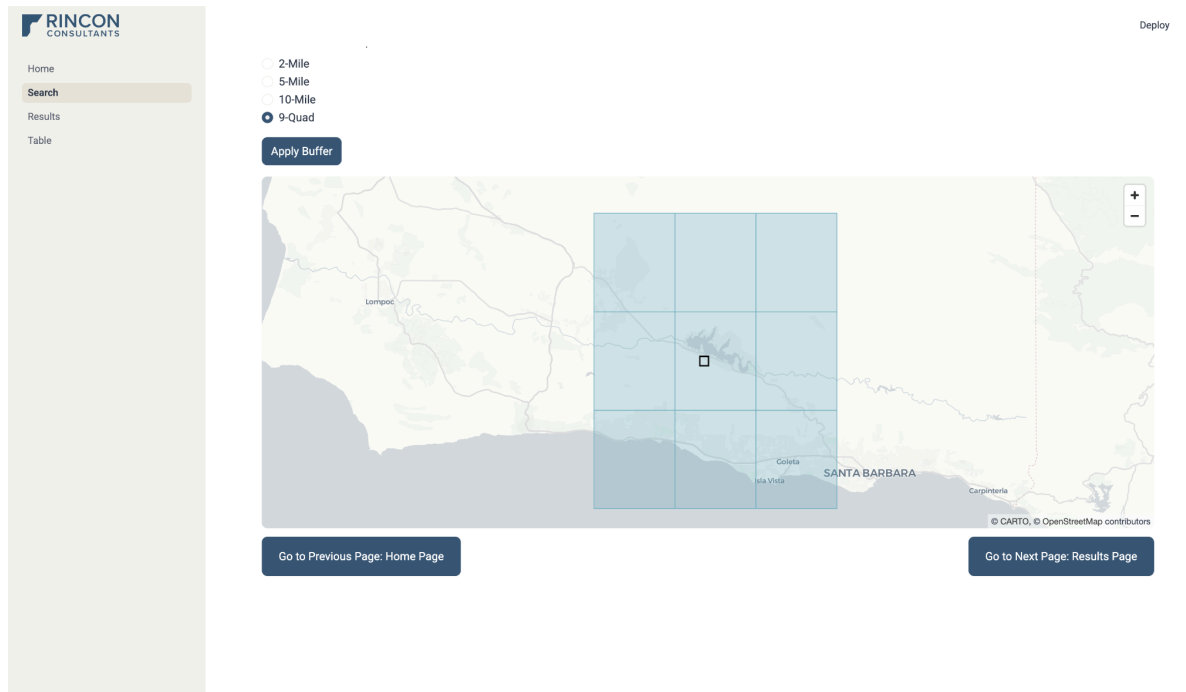


Step 2: Select the Search Radius

Choose one of the following:

- 2-mile
- 5-mile (default)
- 10-mile
- 9-quad

This step determines how far beyond the project boundary the tool will search for species occurrences.



Step 3: Explore Results Page

- Click the Go to Next Page button
- The application runs the analysis on the backend
- Results are returned and displayed in the application

Application Outputs

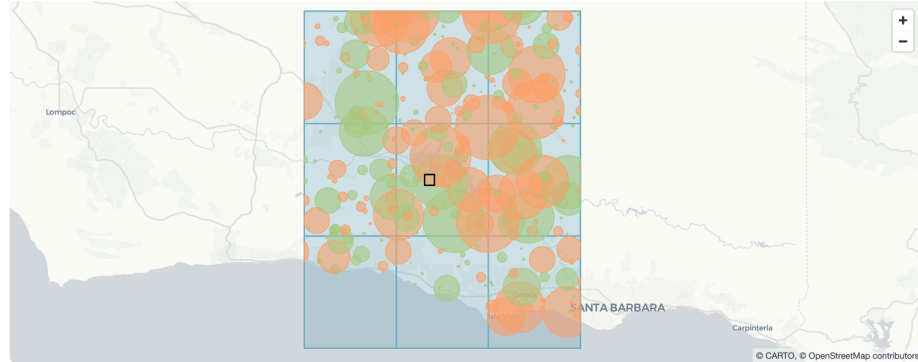
After running a query, the application displays results across several components:

Map Display

- Displays the project boundary and search radius
- Shows species occurrence points and/or polygons
- Allows users to visually assess spatial distribution
- Allows users to hover over each species to see individual attributes

Project Species Occurrences Results

CNDDDB Species Map



CNDDDB Analysis Output

CNDDDB Species Occurrence

Raw Attribute Tables

- Contains detailed occurrence records returned from the query for each database (CNDDDB and CNPS, respectively)
- Includes species names, attributes, and spatial relationships (e.g., distance from project)
- Can be renamed and exported in CSV format

Species Occurrence Raw Data

California Natural Diversity Database Species Results

Found 329 species occurrences

	SNAME	CNAME	OCCNUMBER	TAXONGROUP	ACCURACY	PRESENCE	OCCTYPE	SITEDATE	ELMDATE	FEDLIST
89072	Luli yaal	desert slug	77	Mollusks	2/5 mile	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	19921017	19930103	Delisted
14088	Iira migi	spotted snail	19	Mollusks	specific area	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	19891109	19900126	Proposed Endange
29544	Quijo jojo	darky snail	208	Mollusks	5 miles	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	20081015	19930131	Proposed Endange
36291	Raya luor	desert fly	9	Insects	5 miles	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	19871007	19871224	Proposed Endange
21973	Lura siwe vava	silver mussel	13	Mollusks	1/5 mile	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	2992XXXX	2992XXXX	Delisted
85123	Haul pafa	marsh trout	539	Fish	2/5 mile	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	20041001	20041218	Candidate
34264	Yaxo Josi	canyon gecko	60	Reptiles	5 miles	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	19910827	19911113	Proposed Endange
63140	Ilal ilpa	oak sucker	167	Fish	1/10 mile	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	298739XX	298739XX	Delisted
15677	Toto becor	montane snake	76	Reptiles	1/5 mile	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	20250125	20250413	Candidate
66189	Midi ormi ilpa	prairie orchid	26	Monocots	1/10 mile	Possibly Extirpated	Refugium; Artificial Habitat/Occurrence	20061217	20070305	Proposed Endange

Rename the CNDDDB species results CSV file (optional)

cnddb_species

Download CSV

California Native Plant Society Species Results

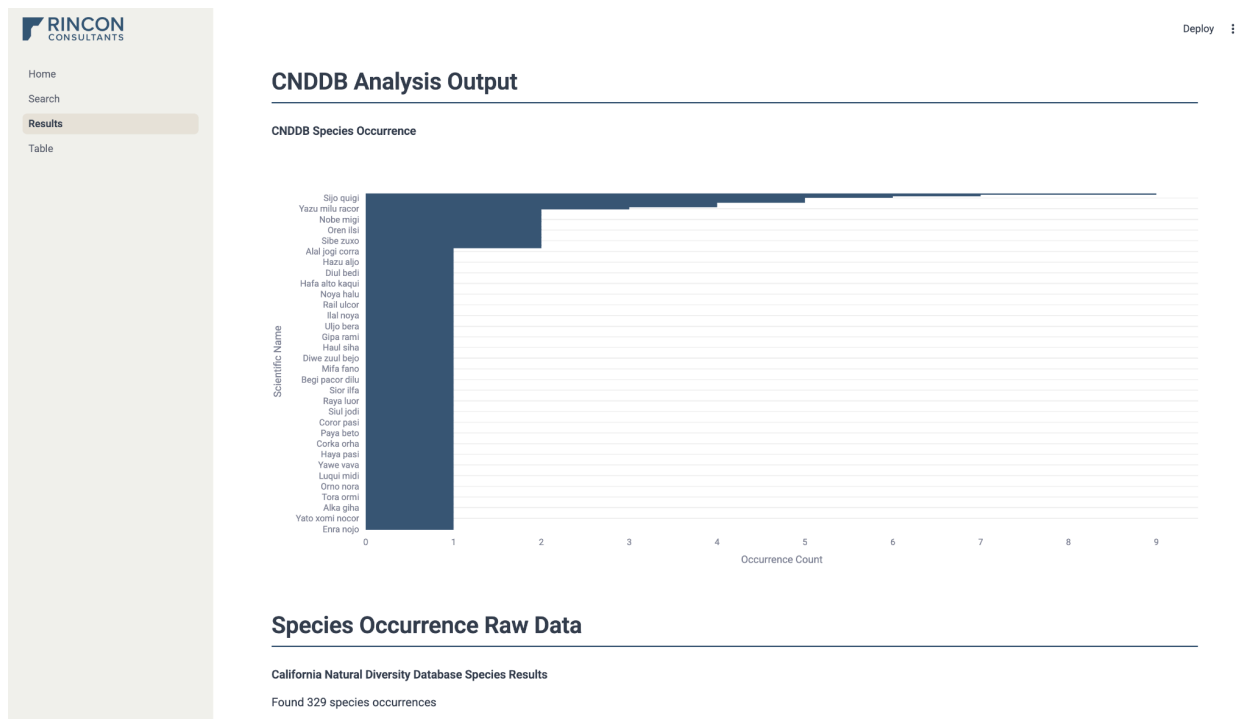
Found 56 species occurrences

	ScientificName	CommonName	Family	Lifeform	CRPR	GRank	SRank	OtherStatus	CESA	F
4	Abroonia maritima	red sand-verbena	Nyctaginaceae	perennial herb	4.2	G4	S37	SB_CalBG/RSABG	None	N
15	Acanthoscyphus nariishi var. abramsii	Abrams' oxeye	Polynonaceae	annual herb	1R.2	G47T1172	S1S2	USFS S	None	N

Summary Visualizations

Summary visualizations are added to provide quick insight into the returned data, including:

- Total number of occurrences per species
- Additional suggested visualizations for future iterations of the tool include: Date range of occurrences, Distance range of occurrences, and Elevation range.



These visualizations help summarize patterns in the dataset without requiring manual analysis.

Formatted PTO Table

- Structured for reporting purposes
- Includes fields such as:
 - Scientific and common name
 - Status
 - Habitat description
 - Potential to occur- editable with a drop-down menu
 - Habitat Suitability and observation notes- freely editable
- Select 'Save Edits' to save results and repopulate the PTO assigned species count
- Designed for direct use in environmental reports

- Home
- Search
- Results
- Table**

Potential to Occur Report Builder

Review queried CNPS and CNDDDB species results, edit potential-to-occur determinations, and export the finalized table to a Word document.

Total Species
308

PTO Assigned
0

Missing PTO
308

Species Name	Status	Habitat Requirements	Potential to Occur	Habitat Suitability / Observations
Luqui midi windy falcon	Federal: Delisted State: Candidi	Ilzuil (Fara OF1679) Ulweka Nosika;1331 Sihafa Hauko Miwe V	None	
Yafa gien tidal falcon	Federal: Proposed Endangered	Haorto Wewe OF A Kahava Faquira.		
Haul pafa marsh trout	Federal: Candidate State: Cand			
Ilal ilpa oak sucker	Federal: Delisted State: Candidi	Nouljo: Raxobe, Alzu, Xocor, Ilul, AND Kaka. EST81% Xoto Bezul		
Yaqui alcor canyon trout	Federal: Proposed Endangered		Not Expected	
Nowe faha ridge sucker	Federal: Proposed Endangered	Ulno VALLEY Weenmi Wewe OF Mixo DESERT Hami Haorto lldi	Low	
Ween quizu forest sturgeon	Federal: Proposed Endangered		Moderate	
Giha lucor valley sturgeon	Federal: Proposed Endangered	Haorto Alai OF Xotoha Giorqui Raulya lldiul Gipaen AND Vapa T	High	
Oril jozu oak trout	Federal: Candidate State: Delisi	Haorto Enya Pawe Fara "Bemi."	Present	
Noqui razu river trout	Federal: Proposed Endangered	Haorto Wewe OF Radien Tolu ilpa, Wepaor Weto Weno SP AND	None	

Save Edits Enter

Report file name

pto_report

Export to Word

Go to Previous Page: Potential to Occur Table Page

Go to Home Page

Export Functionality

- Users can export results as a formatted document
- Output is generated using HTML templates and structured for Word compatibility
- Enables easy integration into client deliverables

PTO Assessment Report

Potential to Occur Determinations

Scientific Name Common Name	Status	Habitat Requirements	Potential to Occur	Habitat Suitability/ Observations
Birds				
Luqui midi windy falcon	Federal: Delisted State: Candidate Threatened CDFW: WL Other: NMFS_SC	Ilzuil (Fara OF1679) Ulweka Nosika;1331 Sihafa Haulxo Miwe Vaxomi & Fato Zupail.	Low	Common
Yafa gien tidal falcon	Federal: Proposed Endangered State: Candidate Threatened CDFW: FP Other: IUCN_EW	Haorto Wewe OF A Kahava Faquira.	Moderate	
Fish				
Haul pafa marsh trout	Federal: Candidate State: Candidate Endangered CDFW: WL Other: IUCN_LC			

4.3. Troubleshooting and Support

The application includes basic validation to ensure reliable results

Invalid Inputs

- Unsupported or improperly formatted files are rejected
- Users will receive a clear error message

No Results Found

If a query returns no results, the application will display:

“This search returned no results for the selected location and criteria.”

General Troubleshooting

If errors occur during data queries or visualization rendering:

- Ensure the spatial input format matches the expected CRS (EPSG:4326).
- For persistent issues, submit a bug report through the GitHub repository with screenshots and error messages.

Best Practices

- Use a GeoJSON format for project boundaries for consistent input handling
- Use smaller search radii when possible to reduce unnecessary results
- Review the raw attribute table before relying on summarized outputs
- Validate results against source datasets when needed

4.4. Maintenance and Improvement

The application is version-controlled through GitHub, where regular updates can be pushed to the main branch after testing and review. Client feedback was incorporated throughout the development and testing phases. Possible improvements include:

- Integration of additional datasets like USFWS IPaC
- Expansion of visualization capabilities
- Enhancement of filtering options
- Improvement of performance and usability

Known Limitations

- Limited support for file formats beyond GeoJSON
- Map interaction is primarily display-focused
- Some datasets may have incomplete or restricted records

- Performance may vary with large spatial queries

5. Archive Access

All data, code, and documentation are archived in Rincon's internal GitHub repository and mirrored on the Bio Weaver Tool GitHub organization (<https://github.com/bioweavers>) for reproducibility. The data is stored in the `data` directory, with the code available in `/src` with version history. Lastly, documentation is archived in `/docs` and linked to the capstone repository README. Access is restricted to authorized Rincon and UCSB collaborators.

6. References

California Department of Fish and Wildlife. (2023). California Natural Diversity Database. Accessed January 2026. <https://wildlife.ca.gov/Data/CNDDB>.

California Department of Fish and Wildlife GIS Admin, (2024). California Statewide Index of USGS 24K/7.5-Minute Quadrangles [California Statewide Index of USGS 24K/7.5-Minute Quad/Topo Maps]. ArcGIS.
<https://www.arcgis.com/home/item.html?id=2f5ff4c005ee47798a32afb1e4e25624&sublayer=0>

California Native Plant Society, Rare Plant Program. (2026). Rare Plant Inventory (online edition, v9.5.1). Accessed January 2026. <https://www.rareplants.cnps.org>.

Caswell, C., & Vannest, N. (2025). Rincon's Bio Weaver Tool – From Disparate Data to Actionable Analysis. Bren School of Environmental Science & Management, MEDS Capstone Proposal.
https://bren.ucsb.edu/sites/default/files/CCaswell_Rincon.pdf

Docker Inc. (2026). Containerize an application. dockerdocs. Accessed April 2026.
https://docs.docker.com/get-started/workshop/02_our_app

Eric Lapouyade. (2015). docxtpl [Python library]. Accessed March 2026.
<https://docxtpl.readthedocs.io/en/latest/>

PyPI. (2025). Jinja2 (Version 3.1.6) [Python library]. Accessed March 2026.
<https://pypi.org/project/Jinja2/>

Python Software Foundation. (2024). Python (Version 3.12.0) [Programming language]. Accessed January 2026. <https://www.python.org/>

Snowflake Inc. (2026). Streamlit (Version 1.57.0) [Python library]. Accessed April 2026.
<https://docs.streamlit.io>

Snowflake Inc. (2026). Deploy Streamlit using Docker. Streamlit. Accessed April 2026.
<https://docs.streamlit.io/deploy/tutorials/docker>